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**Term Project: AI and Society:
Ethical Impact Analysis**

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Introduction

Artificial Intelligence affects many areas of modern society, including work, education, healthcare, research, communication, and the environment. This report analyzes AI's benefits and risks using three ethical frameworks: consequentialism, deontological ethics, and virtue ethics. These frameworks help evaluate AI based on its outcomes, its respect for rights and duties, and the values shown by developers and users. Our group divided the work into four topics. Alejandro A. Roberts Quintana analyzed AI's environmental impact, Jesús Y. Cabán Feliciano analyzed deepfakes and misinformation, Victor A. Gerena Hilerio analyzed AI's positive impacts in healthcare, education, and research, and Nollan N. Rivera Febus analyzed AI's effects on employment. After completing our individual research, the group compared the findings to present a balanced ethical evaluation of AI's development and use.

Overview of AI's Societal Impacts

Artificial Intelligence affects many areas of society, including communication, education, healthcare, research, media, and employment. This report focuses on four main impacts: environmental sustainability, potential misuse, positive applications, and labor market changes. AI creates environmental concerns because large models and data centers require significant electricity and water. It can also be misused through deepfakes, impersonation, and misinformation, which can harm trust and personal security. At the same time, AI has positive applications in healthcare, education, and scientific research by helping doctors, students, and researchers work more efficiently. Finally, AI affects employment by automating tasks, increasing productivity, and changing the skills workers need. Together, these areas show that AI offers important benefits, but also creates ethical challenges related to sustainability, fairness, trust, privacy, and economic security.

Energy Consumption and Sustainability Challenges of AI - Alejandro A. Roberts Quintana

Over the years we have seen multiple industries affect the environment in many different ways, from pollution of air and water, to exploitation of limited natural resources. One of the Artificial Intelligence industry's biggest challenges is to produce sufficient energy for the large AI models and datacenters.. While Artificial Intelligence continues to grow rapidly, environmental cost and energy production becomes difficult to provide. For example, The training process for an LLM can consume thousands of megawatt hours of electricity and emit hundreds of tons of carbon, leading to the evaporation of an astonishing amount of freshwater into the atmosphere for data[1]. Similarly, Because of its benefits of work implementations and productivity, generative AI has enabled rapid deployment of powerful models. But According to Zewe(2025), "The computational power required to train generative AI models that often have

billions of parameters such as OpenAI's GPT-4, can demand a staggering amount of electricity, which leads to increased carbon dioxide emissions and pressures on the electric grid"[2]. The question then becomes: How can the AI industry provide energy and maintain an ethical standard, protecting the environment.

After analysing it in a Consequentialism point of view, and asking: is the carbon emissions and water/resources consumption justified by AI's benefits? It is proven that AI has improved productivity, research, healthcare, automation and scientific development. But at the same time it is affecting the environment we live in, and resources we use everyday. Zewe explains this by stating that "Beyond electricity demands, a great deal of water is needed to cool the hardware used for training , deploying, and fine-tuning generative AI models, which can strain municipal water supplies and disrupt local ecosystems"[2]. AI's environmental impact may be ethically acceptable only if its overall benefits to society outweigh the environmental damage it causes. Analysing from a Deontologist, we can ask: Do companies have a duty to minimize these harms? Companies developing AI have an ethical duty to reduce environmental harm regardless of the benefits their technologies provide. Organizations should act responsibly by improving energy efficiency, increasing transparency regarding resource consumption, and adopting cleaner energy practices. Girolimon explains this in an interview with the vice president of artificial intelligence at SNHU, Dr. Robert MacAusian, where he states that "Organizations must incorporate ethical governance into their AI strategies from the beginning"[3]. Additionally, Girolimon states that while other services like streaming and transportation "have their own impact in the environment, AI's growth is leading to especially dramatic impacts...spurring calls for more procedures surrounding measuring, disclosing and reducing AI's environmental consequences"[3]. From a Virtue Ethics perspective, we can ask: Does responsible AI development reflect sustainability and good character? Responsible AI development should reflect virtues such as sustainability, accountability, honesty, and supervision of natural resources. A virtuous company would seek environmentally conscious solutions like more efficient models and sustainable infrastructure. Companies should prioritize responsible creation rather than growth at any cost.

Overall the ethical framework follows the main idea, Although AI has accomplished incredible achievements and has many applications, it does not outweigh the environmental impact that it is causing, therefore making the AI industry responsible to find ethical solutions to reducing the environmental impact and not prioritize innovation over sustainability.

Potential for Misuse: Deepfakes, Impersonation, and Misinformation- Jesús Y. Cabán Feliciano

One important ethical issue related to artificial intelligence is the possibility of misuse through deepfakes, impersonation, and misinformation. Generative AI tools such as ChatGPT,

DALL-E, Gemini, and similar systems can create realistic text, images, audio, and video. These tools can be useful in education, research, accessibility, and creative work, but the same technology can also be used to deceive people. This becomes a serious concern when AI-generated content is realistic enough to harm someone's reputation, influence public opinion, commit fraud, or use a person's identity without permission. A clear example of this problem is AI voice impersonation. In 2024, the Federal Communications Commission declared AI-generated voices in robocalls illegal because of concerns that voice-cloning technology could be used for scams and voter deception [2]. One reported case involved robocalls that imitated President Joe Biden's voice and discouraged voters from participating in the New Hampshire primary [2]. Another case occurred in Hong Kong, where criminals used deepfake video technology to impersonate company executives and convince an employee at Arup to transfer about \$25 million [3]. These examples show that AI misuse is not only a future risk. It is already causing political, financial, and personal harm.

From a consequentialist perspective, the main question is whether the consequences of this technology produce more benefit than harm. The course reading explains that consequentialism evaluates actions based on their outcomes [1]. In the case of AI misuse, the negative consequences are significant. Deepfakes and misinformation can spread false beliefs, damage public trust, manipulate voters, and help criminals commit scams. The World Economic Forum identified misinformation and disinformation as major global risks in 2024, which shows that this issue can affect society as a whole, not only individual victims [4].

From a deontological perspective, the misuse of AI is wrong because it violates duties such as honesty, consent, privacy, and respect for human dignity. The ethics reading explains that deontological ethics focuses on moral rules, rights, and obligations, even when the outcome might seem beneficial to someone [1]. A deepfake that uses a person's face or voice without permission treats that person as a tool for deception. This is especially serious in cases of non-consensual synthetic intimate imagery, where a person's identity and body are exploited without consent. A 2024 study across ten countries found that non-consensual deepfake pornography is already a measurable problem, both in victimization and perpetration [5].

From a virtue ethics perspective, this type of AI misuse shows poor moral character. The course reading explains that virtue ethics focuses on traits such as honesty, fairness, compassion, and responsibility [1]. Creating or spreading deepfakes for deception reflects dishonesty and a lack of concern for the people affected. If society becomes used to this type of content, people may become less trusting and less committed to truth. For this reason, developers, companies, and users should promote responsible habits, such as labeling synthetic content, adding safeguards, limiting harmful uses, and responding seriously when victims report abuse.

The three ethical frameworks reach a similar conclusion, but each one emphasizes a different reason. Consequentialism focuses on the social harm caused by AI misuse. Deontology

focuses on the violation of consent, truth, and personal rights. Virtue ethics focuses on the irresponsible character shown by deceptive uses of AI. Overall, deontological ethics seems to be the strongest framework for this issue because the central problem is the violation of consent and honesty. Even before considering the larger consequences, impersonating someone or creating a fake version of them without permission is already ethically wrong. For that reason, AI should not be rejected completely, but its development and use should include strong safeguards against deception, impersonation, and non-consensual use.

Positive Impacts of Artificial Intelligence - Victor A Gerena Hilerio

Artificial Intelligence has strong positive impacts in healthcare, education, and scientific research when used responsibly. Modern AI systems, especially deep learning and transformer-based models, can analyze large amounts of data, recognize complex patterns, and support human decision-making. These benefits can be evaluated through consequentialism, deontological ethics, and virtue ethics. Consequentialism focuses on outcomes, deontological ethics focuses on duties and rights, and virtue ethics focuses on responsible values such as fairness, honesty, and compassion [1]. In medical diagnostics, AI can help doctors detect diseases earlier and more accurately by analyzing medical images, laboratory results, and patient data. The U.S. Food and Drug Administration already maintains a public list of AI-enabled medical devices, showing that AI is becoming part of regulated healthcare practice [3]. From a consequentialist perspective, this is valuable because earlier diagnosis can reduce suffering and save lives. However, these benefits are ethical only if AI systems work accurately for different populations and do not increase health disparities. From a deontological view, medical AI must respect privacy, informed consent, and accountability. Virtue ethics adds that healthcare institutions should use AI with compassion, humility, and responsibility. AI also has positive impacts in education. Generative AI and intelligent tutoring systems can explain concepts, create examples, translate material, and provide personalized feedback. This can help students who need extra support outside the classroom. UNESCO has recognized the importance of generative AI in education while emphasizing responsible and human-centered use [5]. Consequentialism supports educational AI when it improves learning and access, but it becomes harmful if students use it to cheat or avoid learning. Deontological ethics emphasizes academic honesty, privacy, and integrity. Virtue ethics reminds students and teachers that AI should encourage curiosity, discipline, and deeper understanding. In scientific research, AI can accelerate discovery by analyzing large datasets, identifying patterns, and helping researchers form hypotheses. AlphaFold is a major example because it achieved a breakthrough in predicting protein structures, which has applications in biology, medicine, and drug discovery [6]. From a consequentialist perspective, this is ethical when it produces knowledge that benefits society. From a deontological perspective, researchers still have duties of transparency, honesty, and verification. Virtue ethics emphasizes humility and responsibility, meaning scientists should treat AI as a tool for discovery, not a replacement for careful human judgment.

Overall, AI has strong positive potential in medicine, education, and scientific research. It can help diagnose disease earlier, make learning more accessible, and accelerate scientific progress. However, these benefits are ethical only when AI is used with fairness, transparency, privacy, and accountability. Therefore, AI should continue to be developed as a tool that supports human dignity and human expertise.

Impact on Employment and Labor Markets-Nollan Rivera

Artificial Intelligence (AI) is transforming labor markets around the world by automating repetitive tasks, improving productivity, and reducing operational costs in industries such as manufacturing, transportation, finance, healthcare, and customer service. Recent advances in generative AI, including systems such as ChatGPT and Gemini, have expanded automation into areas that were previously associated with highly skilled workers, including writing, programming, and data analysis [1]. Despite these benefits, AI also raises concerns regarding job displacement. Many occupations that rely on routine cognitive or manual tasks face a higher risk of automation. According to the World Economic Forum, while AI is expected to create millions of new jobs, it is also projected to displace millions of existing positions, forcing workers to adapt through education, retraining, and lifelong learning [2]. As a result, labor markets are experiencing significant changes that create both opportunities and challenges for workers.

After analyzing this issue from a consequentialist perspective, the main question becomes whether the benefits of AI outweigh its negative consequences. AI can increase productivity, improve economic growth, and allow workers to focus on more complex and creative tasks. However, large-scale job displacement can also create unemployment, income inequality, and economic insecurity. The International Labour Organization explains that the effects of generative AI on employment will depend heavily on how organizations implement these technologies and whether workers are given opportunities to adapt to changing job requirements [3]. Therefore, a consequentialist would support AI adoption only. From a deontological perspective, the focus shifts from outcomes to duties and moral responsibilities. Companies should not replace workers simply because AI can perform certain tasks at a lower cost. Organizations have a responsibility to treat employees fairly, communicate honestly about automation plans, and consider the impact that technological changes may have on workers and their families. Russell and Norvig emphasize that AI systems should be developed and used in ways that align with human values and ethical responsibilities, rather than focusing exclusively on efficiency and profit. From a virtue ethics perspective, responsible AI development should reflect virtues such as fairness, responsibility, honesty, and compassion. A virtuous organization would seek to balance innovation with concern for its workforce by investing in retraining programs and creating opportunities for employees to develop new skills. Instead of viewing workers as replaceable resources, ethical leaders should demonstrate accountability and

commitment to the long-term well-being of both employees and society. This approach promotes trust and encourages the responsible integration of AI into the workplace.

Our group considers consequentialism to be the most appropriate ethical framework for evaluating AI's impact on employment and labor markets. This issue largely depends on balancing the benefits of increased productivity, innovation, and economic growth against the risks of unemployment and inequality. While deontological ethics and virtue ethics provide valuable perspectives regarding worker rights and organizational responsibility, consequentialism offers the most comprehensive approach because it directly evaluates the overall impact of AI on society as a whole.

Group Deliberation Process

The group reviewed four AI topics: environmental impact, deepfakes and misinformation, positive impacts, and employment. Each was analyzed using consequentialism, deontological ethics, and virtue ethics. Consequentialism focused on AI's benefits and harms. AI can improve healthcare, education, research, and productivity, but it can also cause environmental damage, job loss, misinformation, and unequal access. Deontological ethics emphasized rights and duties such as privacy, consent, honesty, and accountability. The group agreed that AI should not violate these principles, even when it provides benefits. Virtue ethics focused on responsible behavior and values such as fairness, honesty, compassion, and sustainability. It highlighted the importance of ethical decision-making by developers and users. The group noted conflicts between innovation and sustainability, productivity and worker protection, and creativity and misuse. Overall, the frameworks complement each other: consequentialism evaluates outcomes, deontology protects rights, and virtue ethics promotes responsibility.

Conclusion

Artificial Intelligence offers major benefits but also creates ethical challenges. It can improve healthcare, education, research, and productivity while also contributing to environmental concerns, job displacement, misinformation, privacy issues, and inequality. The group concluded that AI should continue to be developed with strong ethical safeguards. Its value depends on how it is designed and used. Consequentialism supports AI when benefits outweigh harms, deontology requires respect for rights and duties, and virtue ethics encourages fairness, responsibility, and concern for the common good. AI should support human expertise rather than replace human responsibility. Transparency, privacy protection, accountability, worker protections, and environmental awareness are essential to ensure AI benefits society fairly.

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Appendix A: Group Meeting Records

A.1 First Group Meeting

Date: [5/25/2026]

Location: [Teams]

Members present: Alejandro A. Roberts Quintana, Jesús Y. Cabán Feliciano, Victor A. Gerena Hilerio, Nollan N. Rivera Febus

Minutes of First Meeting

The group reviewed the assignment requirements and identified the main sections of the report. Four AI topics were selected: environmental impact, deepfakes and misinformation, positive impacts, and employment effects.

Assignments:

- Alejandro A. Roberts Quintana — Environmental impact
- Jesús Y. Cabán Feliciano — Deepfakes and misinformation
- Victor A. Gerena Hilerio — Positive impacts of AI
- Nollan N. Rivera Febus — Employment and labor markets

The group agreed to apply consequentialism, deontological ethics, and virtue ethics in each section and use credible sources.

Decisions Made

1. Focus on four AI impact areas.
2. Assign one section to each member.
3. Apply all three ethical frameworks.
4. Include individual reflections.
5. Use IEEE references.
6. Combine all sections into one report.

A.2 Final Group Meeting

Date: [5/29/2026]

Location: [Teams]

Members present: Alejandro A. Roberts Quintana, Jesús Y. Cabán Feliciano, Victor A. Gerena Hilerio, Nollan N. Rivera Febus

Minutes of Final Meeting

The group reviewed all sections and compared ethical conclusions. Members discussed how the frameworks addressed AI's benefits, risks, rights, and responsibilities.

The group agreed that consequentialism helps assess outcomes, deontology protects rights and duties, and virtue ethics promotes responsible values. They concluded that AI should continue to develop but with transparency, accountability, fairness, privacy protection, and environmental responsibility.

Decisions Made

1. AI has significant benefits and risks.
2. AI development should continue with safeguards.
3. Emphasize transparency, accountability, sustainability, fairness, and privacy.
4. Use all three ethical frameworks.
5. Finalize the conclusion and appendix sections.

Personal Paragraph - Alejandro A. Roberts Quintana

I contributed to the group by researching the environmental impact of Artificial intelligence, focusing on energy consumption, carbon emissions, and resource usage within AI models and datacenters. Using the ethical framework given, I analyzed the topic and learned that although AI offers a lot of benefits in many fields and daily life, it also presents significant environmental challenges that require more supervision and ethical guidelines in the industry.

Additionally, I learned that by applying the ethical frameworks, I can work my answers by viewing different perspectives and tackle different interpretations of the main idea.

Personal Paragraph - Jesús Y. Cabán Feliciano

For this assignment, I learned how the misuse of artificial intelligence can create serious ethical problems, especially through deepfakes, impersonation, and misinformation. Before working on this topic, I understood that AI could generate realistic content, but I did not fully realize how this technology could be used to deceive voters, commit financial fraud, or violate a person's consent by using their face or voice without permission. My contribution was researching and writing the section on the potential misuse of AI and applying the three ethical frameworks: consequentialism, deontological ethics, and virtue ethics. Through this work, I learned that deontological ethics is especially important for this issue because AI misuse often violates basic duties such as honesty, consent, and respect for others.

Personal Paragraph - Victor A Gerena

Through this assignment, I learned that Artificial Intelligence should not be analyzed only as a technological advancement, but also as a tool that affects real people and society. My topic focused on the positive impacts of AI in medical diagnostics, education, and scientific research. I contributed by explaining how AI can help doctors detect diseases earlier, support students with personalized learning, and accelerate scientific discoveries such as protein structure prediction. I also applied the three ethical frameworks to show that these benefits must still respect privacy, fairness, transparency, and accountability. Overall, this assignment helped me understand that responsible AI should support human expertise and improve society without replacing human judgment or ignoring ethical responsibilities.

Personal Paragraph - Nollan Rivera

Through this assignment, I learned that Artificial Intelligence can have both positive and negative effects on employment. Before researching this topic, I mainly focused on the technological benefits of AI, but I now better understand the challenges it creates for workers and labor markets. My contribution to the group was researching AI's impact on employment and labor markets and applying the ethical frameworks discussed in class to analyze the issue. This assignment helped me see that the development of AI should not only focus on efficiency and innovation, but also on protecting workers and creating opportunities for people to adapt to technological changes.